

A Novel Approach to Tackling Bullying in Schools: Personality-Targeted Intervention for Adolescent Victims and Bullies in Australia

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Objective: To examine the secondary effects of a personality-targeted intervention on bullying and harms among adolescent victims and bullies.

Method: Outcomes were examined for victims and bullies in the Climate and Preventure study, Australia. Participants completed self-report measures at baseline and four follow-up assessments (6, 12, 24, and 36 months). Thirteen intervention schools ($n = 1,087$) received Preventure, a brief personality-targeted cognitive-behavioral therapy intervention for adolescents with high-risk personality types (hopelessness, anxiety sensitivity, impulsivity, sensation seeking). Thirteen control schools ($n = 1,103$) received health education as usual. Bullying was examined for high-risk victims ($n = 143$ in Preventure schools versus $n = 153$ in control schools) and bullies ($n = 63$ in Preventure schools versus $n = 67$ in control schools) in the total sample. Harms were examined for high-risk victims ($n = 110$ in Preventure schools versus $n = 87$ in control schools) and bullies ($n = 50$ in Preventure schools versus $n = 30$ in control schools) in independent schools.

Results: There was no significant intervention effect for bullying victimization or perpetration in the total sample. In the subsample, mixed models showed greater reductions in victimization ($b = -0.208$, 95% CI -0.4104 to -0.002 , $p < .05$), suicidal ideation ($b = -0.130$, 95% CI -0.225 to -0.034 , $p < .01$), and emotional symptoms ($b = -0.263$, 95% CI -0.466 to -0.061 , $p < .05$) among high-risk victims in Preventure versus control schools. Conduct problems ($b = -0.292$, 95% CI -0.554 to -0.030 , $p < .05$) showed greater reductions among high-risk bullies in Preventure versus control schools, and suicidal ideation showed greater reductions among high-risk female bullies in Preventure versus control schools ($b = -0.820$, 95% CI -1.198 to -0.442 , $p < .001$).

Conclusion: The findings support targeting personality in bullying prevention.

Clinical trial registration information: The CAP Study: Evaluating a Comprehensive Universal and Targeted Intervention Designed to Prevent Substance Use and Related Harms in Australian Adolescents; <http://www.anzctr.org.au/>; ACTRN12612000026820.

Key words: bullying, early intervention, mental health, school-based programs

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Bullying is an important global public health problem that affects millions of people each year, particularly school-age children.¹ In a nationally representative study in Australia, the prevalence of frequent bullying in the past year (every few months or more) among youths 11 to 14 years old was 16% for victimization only, 2% for perpetration only, and 3% for both victimization and perpetration.² The negative impact of bullying during childhood and adolescence is substantial, with long-lasting psychosocial consequences for individuals involved.^{3–7} The harms of bullying are not limited to victimization, with clear evidence that involvement as a victim and/or a bully is associated with

internalizing problems, such as anxiety and depression, and externalizing problems, such as conduct problems and substance use.^{2,6} In fact, adolescents concurrently involved in both forms of bullying, bully-victims, are typically found to experience greater psychological difficulties than either victims or bullies.^{8,9} Furthermore, there is evidence that bullying can be chronic for some adolescents, and this chronic involvement results in greater harms than a discrete period of bullying.¹⁰ Suicide is an area of particular concern for adolescents with chronic bullying involvement and for bully-victims.^{8,11–13} Importantly, recent evidence from twin studies indicates that the association between bullying and psychological

difficulties is a combination of exposure to the bullying itself and preexisting vulnerabilities.^{5,6}

The overwhelming evidence of the negative impacts of bullying highlights the need for schools to implement effective interventions to address bullying behaviors.³ However, more recent research highlights limitations in current bullying intervention, particularly for adolescents.^{14,15} Currently, most bullying interventions are universal interventions delivered to all students regardless of risk; there is a great lack of evidence-based interventions for adolescents involved in bullying (indicated prevention).¹⁵ There is promising evidence for the use of a cognitive-behavioral therapy (CBT) approach to indicated bullying intervention for adolescents; however, these programs are lengthy and can suffer from poor implementation.¹⁶⁻¹⁸ It is imperative that indicated bullying interventions are feasible and focus on reducing additional bullying involvement among victims and/or bullies as well as addressing psychological harm among those involved.¹⁹ Not all adolescents involved in bullying experience distress or harm as a result.²⁰ It is therefore important that interventions target individuals most likely to experience problems as a result of bullying and those most likely to have preexisting psychological problems. Such interventions have the potential to prevent long-lasting mental health problems and to save the lives of the most vulnerable adolescents. Innovative strategies are needed to reduce bullying and the harm experienced by those involved.¹⁹ Furthermore, the overlap of bullying roles underscores the need for bullying interventions that can be used for victims and/or bullies.

A promising innovative approach to indicated bullying prevention is personality-targeted intervention. Children's personalities influence the way they experience, interpret, and respond to the world around them, such as their social interactions and perceptions about themselves and others.²¹ Personality traits are implicated in internalizing and externalizing problems among adolescents.^{22,23} There is also evidence that personality traits predict involvement in bullying victimization and perpetration.^{24,25} Personality is a particularly useful factor to target, as it influences a range of other risk factors for bullying, such as anxiety, social skills, and self and/or other cognitions. Personality also interacts with pertinent environmental factors, such as family and peer relationships; maladaptive personality traits and adverse environmental factors can mutually reinforce each other.²² This is consistent with the view of bullying as a complex interaction between the individual and his or her environment.²⁶

A brief CBT-based personality-targeted intervention, Preventure, has been shown to be effective in reducing internalizing and externalizing problems among adolescents with four personality types shown to increase the risk of

substance use and psychopathology: hopelessness, anxiety sensitivity, impulsivity, and sensation seeking.²⁷ The current study examines whether Preventure is effective in reducing additional bullying involvement among adolescent victims and bullies with these high-risk personality types, as well as reducing harms among these adolescents. Specifically, it is hypothesized that:

1. Bullying victimization and perpetration will be reduced more among adolescent victims and bullies with high-risk personality types in Preventure schools than those in control schools;
2. Suicidal ideation, internalizing problems, and externalizing problems will be reduced more among adolescent victims and bullies with high-risk personality types in Preventure schools than those in control schools.

METHOD

Study Design

The current study examined secondary effects of a personality-targeted group program on bullying, suicidality, internalizing problems, and externalizing problems among victims and bullies in the Climate and Preventure (CAP) study. The CAP study is a cluster randomized controlled trial (RCT) of a comprehensive substance use prevention intervention for adolescents that combined a universal substance use prevention program called Climate Schools with the targeted substance use prevention program, Preventure.²⁸ The CAP study included 26 secondary schools (18 independent and 9 public) in New South Wales and Victoria, Australia. Initially, 27 schools agreed to participate; however, owing to time constraints, one school (assigned to the Climate condition) withdrew after randomization but before completing baseline questionnaires. Of the 2,608 eligible students invited into the study, 2,190 provided consent and completed the baseline survey between February and May 2012. A self-report questionnaire was administered to all students in a classroom setting at baseline and the four subsequent follow-up assessments: post intervention and 12 months, 24 months, and 36 months post baseline. Student responses were linked over time using a unique identification code to ensure confidentiality. Schools were randomized to receive Climate Schools and Preventure, Climate Schools only, Preventure only, or drug education as usual. The CONSORT diagram in Figure S1 (available online) summarizes participant flow and retention rates through the study for each condition. Study retention was high; 36-month follow-up rates for each arm of the CAP study ranged from 64% to 80% (Figure S1, available online), and only 4% of participants completed no follow-

up surveys. The research protocol,²⁸ including informed consent procedures and sample size calculations, was approved by the University of New South Wales Human Research Ethics Committee, the Sydney Catholic Education Office, and the New South Wales Department of Education and Training. The CAP trial is registered with the Australian New Zealand Clinical Trials Registry (ACTRN12612000026820). The substance use outcomes of the CAP study and primary alcohol use outcomes for Preventure have been reported elsewhere.^{29,30}

The current study collapsed the Climate Schools and Preventure arm and Preventure-only arm of the CAP study into the intervention group (Preventure schools, $n = 1,087$) and combined the Climate Schools only and drug education as usual group into the control group (control schools, $n = 1,103$). This decision was made after data collection to provide sufficient power to detect changes in continuous measures. This study compared the outcomes for victims and bullies (classified at baseline) screened as having high-risk personality types (hereby referred to as high-risk victims and high-risk bullies) in Preventure schools versus control schools. Bullying outcomes were assessed for high-risk victims and high-risk bullies in the whole sample, whereas the harms were assessed among high-risk victims and high-risk bullies in the 17 private/independent schools (the harms were not measured in the public schools owing to ethical requirements for public schools). Both victimization and perpetration were assessed for victims and bullies owing to the overlap between these roles, both concurrently and over time.

Measures

Demographic Information. Participants were asked to report sociodemographic information, including age, gender, and country of birth.

Personality. Personality was measured using the Substance Use Risk Profile Scale (SURPS).³¹ The SURPS was originally developed to assess four personality traits relevant to substance misuse: impulsivity, hopelessness, anxiety sensitivity, and sensation seeking.³² The 23-item SURPS was validated using factor analysis on a battery of personality and symptom inventories that assess these personality dimensions.^{31,32} The SURPS has been shown to predict specific patterns of substance use, psychopathology, and bullying involvement.³²⁻³⁴ Anxiety sensitivity is a fear of anxiety-related physical sensations and includes items such as “It’s frightening to feel dizzy and faint” and “I get scared when I’m too nervous.” Hopelessness reflects a tendency toward low mood, feelings of worthlessness, and negative beliefs about oneself, the world, and the

future and includes items such as “I am very enthusiastic about my future” (reverse scored) and “I feel that I’m a failure.” Impulsivity can be described as rapid decision making/action and poor response inhibition and includes items such as “I usually act without stopping to think” and “I often involve myself in situations that I later regret being involved in.” Sensation seeking refers to an elevated need for stimulation and intolerance to boredom and includes items such as “I like doing things that frighten me a little” and “I am interested in experience for its own sake, even if it is illegal.” Previous studies in schools have demonstrated utility in categorizing students scoring 1 SD above their school mean on any of the four personality risk subscales as high-risk, indicating that they are at increased risk of substance use or psychopathology associated with that personality type.³²

Bullying. Bullying prevalence was measured using an amended version of the Revised Olweus Bully/Victim Scale.³⁵ This scale has satisfactory psychometric properties and demonstrated excellent internal consistency in this sample ($\alpha = .9$).³⁶ The bullying questionnaire provided the respondents with a definition of bullying and asked them to indicate how often they had been involved in bullying in the past 6 months (including general bullying victimization and perpetration as well as verbal, relational, and physical victimization and perpetration). Dichotomous variables were created to select the victim and bully groups, based on a definition of fortnightly or more frequent involvement (participants may be in both victim and bully groups). The Revised Olweus Bully/Victim scale was also used to compute total victimization and perpetration scores, by totaling the frequency of verbal, relational, and physical victimization and perpetration, respectively. The total scores were log-transformed owing to skewness; the transformation successfully normalized all variables.

Internalizing Problems. Internalizing problems in the previous 6 months were measured through the five-item emotional symptoms subscale of the Strengths and Difficulties Questionnaire (SDQ) (self-report for youths 11–17 years old), including items such as “I worry a lot” and “I am often unhappy, depressed or tearful.”³⁷ The scale has good psychometric properties.^{38,39} A dichotomous variable was created for the descriptives (using the cutoff of abnormally high scores that identifies likely cases with mental health disorders,³⁷ high versus borderline/normal), and a total score was computed for the outcome measure.

Externalizing Problems. Externalizing problems in the previous 6 months were measured through the five-item conduct subscale of the SDQ described above, including

items such as “I am often accused of lying or cheating” and “I take things that are not mine from home, school or elsewhere.” For descriptive purposes, a dichotomous variable was created (high versus borderline/normal as above), and a total score was computed for the outcome measure.

Suicidality. Suicidal ideation was measured using a question from the Brief Symptom Inventory,⁴⁰ asking how often in the past 6 months the respondent had had “thoughts of ending your life.” A dichotomous variable was created for descriptive purposes (moderately frequent or more versus infrequent or no suicidal thoughts), and a total score was computed for the outcome measure.

Intervention

Preventure is a brief group-based intervention for individuals with personality types shown to be risk factors for substance use and other emotional and behavioral problems among adolescents.³² It has been modified for use among adolescents in Australian schools.⁴¹ The Preventure program includes two 90-minute group sessions per personality profile delivered 1 week apart, based on motivational interviewing and CBT. The first group includes goal-setting, psychoeducation about the personality profile being targeted and common problematic coping behaviors for that personality type, an introduction to the CBT model, and practice in applying the CBT model to their own experiences. In the second group, students receive psychoeducation about unhelpful thinking styles associated with their personality profile and practice identifying and challenging these beliefs to be more able to achieve their goals. The Preventure groups were delivered by trained facilitators (registered psychologists) and cofacilitators (minimum training: bachelor of psychology [honours] degree) and were run during school time. Additional details on the Preventure program have been described elsewhere.^{28,41} Selection into the Preventure intervention was determined according to the student scores on the SURPS. Students in intervention schools that scored 1 SD above their school mean on any of the four personality risk subscales were categorized as high risk and were allocated to the corresponding group (hopelessness = 94; anxiety sensitivity = 150; impulsivity = 103; sensation seeking = 124). A total of 81 groups (162 sessions) were completed between March and November 2012 (predominantly year 8 students; age 12–15 years), with an average of five students per group.

Statistical Analyses

Multilevel mixed-effects analysis for repeated measures was conducted using Stata 15.0 (StataCorp LLC, College

Station, Texas) to account for clustering at the school level and to model change over time.⁴² All models were based on the intention-to-treat principle, using all available measurements from participants and according to their randomly allocated group. Missing data were accommodated based on all available information using maximum likelihood estimation. The effects of the interventions on change in outcomes over time were assessed through group $\times$ time interaction terms. Gender $\times$ intervention effects were nonsignificant for all outcomes except suicidal ideation for bullies; intervention effects were therefore examined separately for male and female bullies for suicidal ideation and in male and female participants combined for all other outcomes (including sex and age as covariates). Effect sizes were calculated by exponentiating the coefficient, to give an indication of intervention effects. All models take into account individual differences at baseline, using baseline measurements as the reference point to estimate participant-specific starting points and change over time from these baseline levels. In line with the CONSORT 2010 guidelines, significance tests of baseline differences between intervention groups were not conducted; however, it should be noted that gender differences were evident (Table S1, available online).

The analyses in the current study were exploratory and devised after data collection for the trial. The original sample size calculations for the RCT, reported in the published trial protocol,²⁸ were based on methods for sample size calculation in longitudinal cluster RCTs for differences in continuous outcomes.⁴³ Whereas power analyses are not traditionally conducted post hoc, a retrospective power analysis provides an indication of whether the current analyses are powered to detect differences between the groups. As the intervention groups were collapsed for this analysis, there were 13 schools per group rather than the 6 on which the original sample size calculations were based on. Estimating 11 high-risk victims per school provided 81% power to detect differences for victims, and estimating 5 high-risk bullies per school provided 48% power to detect differences for bullies. Whereas the focus of the study is on the outcomes for adolescents involved in bullying at baseline, multilevel mixed-effects analysis for repeated measures was also conducted to examine the bullying outcomes for the total sample to assess possible population effects.

RESULTS

Characteristics of Sample

The mean age of the students in the CAP sample at baseline (N = 2,190) was 13.3 years, and 57% of the students were boys. One quarter of the students were in public schools

($n = 554$), and the remainder were in private/independent schools ($n = 1,636$). Half of the students were in Preventure schools ($n = 1,087$), and half of the students were in control schools ($n = 1,103$). Most (93%) of the students were born in Australia, 4% were born in other English-speaking countries, and 3% were born in non-English-speaking countries. Whereas the victims and bullies in Preventure schools were similar to those in the control schools in regard to their age, personality, and bullying involvement, there was a much higher proportion of male participants in the Preventure schools (80.5%) compared with the control schools (34.6%) (Table S1, available online). Consequently, effects are examined separately for male and female participants, where there is a significant effect of gender, and gender was controlled for in the remaining analyses.

One quarter of the students were classified as victims at baseline, and almost 1 in 10 (9%) were classified as bullies at baseline. Both frequent bullying victimization and frequent bullying perpetration were reported by 6% of the sample. Male participants were more likely than female participants to be victims (28% of male participants were victims versus 24% of female participants: $\chi^2_1 4.570, p = .033$) and bullies (12% of male participants were bullies versus 5% of female participants: $\chi^2_1 20.925, p = .000$). Victims and bullies were more likely than uninvolved students to be screened as having high-risk personality types (37% of uninvolved students versus 57% of victims: $\chi^2_1 62.629, p = .000$; 37% of uninvolved students versus 70% of bullies: $\chi^2_1 47.409, p = .000$). Among the private/independent school students, approximately 1 in 10 students reported frequent suicidal ideation (9%), emotional symptoms in the abnormally high range (8%), and conduct problems in the abnormally high range (12%). Both victims and bullies were significantly more likely than uninvolved students to report frequent suicidal ideation (5% of uninvolved students versus 22% of victims: $\chi^2_1 96.150, p = .000$; 5% of uninvolved students versus 23% of bullies: $\chi^2_1 55.497, p = .000$), high emotional symptoms (5% of uninvolved students versus 18% of victims: $\chi^2_1 63.127, p = .000$; 5% of uninvolved students versus 20% of bullies: $\chi^2_1 43.971, p = .000$), and high conduct problems (8% of uninvolved students versus 20% of victims: $\chi^2_1 39.640, p = .000$; 8% of uninvolved students versus 37% of bullies: $\chi^2_1 93.910, p = .000$).

Preventure Outcomes

Bullying Victimization and Perpetration. In the total sample, there was no significant intervention effect for bullying victimization ($b = -0.02$, 95% CI -0.08 to 0.03 , $p > .05$) or perpetration ($b = -0.02$, 95% CI -0.07 to

0.02 , $p > .05$). High-risk victims in Preventure schools reported a significantly greater reduction in bullying victimization over the 3-year study period than high-risk victims in control schools (Table 1, Figure 1A). There was no significant difference in perpetration over time for high-risk victims in Preventure versus control schools (Table 1, Figure 1B). There was no significant difference in reported bullying perpetration or bullying victimization over the study period for high-risk bullies in Preventure schools compared with high-risk bullies in control schools (Table 1, Figure 2A, B).

Internalizing Problems, Externalizing Problems, and Suicidal Ideation. High-risk victims in Preventure schools reported significantly fewer emotional symptoms and less suicidal ideation over time compared with high-risk victims in control schools (Table 1, Figure 3A, C). There was no significant difference in conduct problems over time between high-risk victims in Preventure schools and high-risk victims in control schools (Table 1, Figure 3B). High-risk bullies in Preventure schools reported significantly fewer conduct problems over time compared with high-risk bullies in control schools (Table 1, Figure 4B). High-risk female bullies in Preventure schools reported significantly less suicidal ideation than high-risk female bullies in control schools (Table 1, Figure 4C), but there was no difference in suicidal ideation for high-risk male bullies in Preventure schools compared with high-risk male bullies in control schools (Table 1, Figure 4D). There was no significant difference in emotional symptoms over time among high-risk bullies in Preventure schools compared with high-risk bullies in control schools (Table 1, Figure 4B).

DISCUSSION

This study aimed to examine the secondary effects of Preventure, a brief personality-targeted CBT group intervention, on reducing additional bullying and harms among victims and bullies in secondary schools in Australia. High-risk victims in schools that were randomized to the Preventure intervention reported significant reductions in their rates of victimization, suicidal ideation, and emotional problems over the 3-year study period. To put this in context, victimization scores were 19% lower, emotional symptom scores were 23% lower, and suicidal ideation scores were 12% lower among high-risk victims in Preventure schools compared with high-risk victims in control schools. In addition, high-risk bullies in Preventure schools showed significant reductions in conduct problems (male and female participants) and suicidal ideation (female participants only).

TABLE 1 Change in Bullying Victimization, Bullying Perpetration, Emotional Problems, Conduct Problems, and Suicidal Ideation Over Time Among High-Risk Victims and Bullies in Preventure Schools Compared With Those in Control Schools

	Linear Change: Baseline to 36 Months			Exp(b)
	b	95% CI	p	
All schools (N = 26)				
High-risk victims (n = 296) ^a				
Victimization	−0.208	−0.414 to −0.002	< .05	.812
Perpetration	−0.108	−0.277 to 0.062	> .05	.898
High-risk bullies (n = 130) ^b				
Perpetration	−0.027	−0.345 to 0.291	> .05	.973
Victimization	−0.118	−0.461 to 0.224	> .05	.889
Private/independent schools (n = 17)				
High-risk victims (n = 197) ^c				
Emotional symptoms	−0.263	−0.466 to −0.061	< .05	.769
Conduct problems	−0.100	−0.253 to 0.053	> .05	.905
Suicidal ideation	−0.130	−0.225 to −0.034	< .01	.878
High-risk bullies (n = 80) ^d				
Emotional symptoms	−0.202	−0.525 to 0.122	> .05	.817
Conduct problems	−0.292	−0.554 to −0.030	< .05	.747
Suicidal ideation ^e				
Female students	−0.820	−1.198 to −0.442	< .001	.440
Male students	−0.175	−0.382 to 0.0432	> .05	.839

Note: ^a143 in Preventure schools, 153 in control schools.

^b63 in Preventure schools, 67 in control schools.

^c110 in Preventure schools, 87 in control schools.

^d50 in Preventure schools, 30 in control schools.

^eSignificant intervention × time × gender interaction, so suicidal ideation was examined separately for female and male participants (gender included as a covariate in all other models).

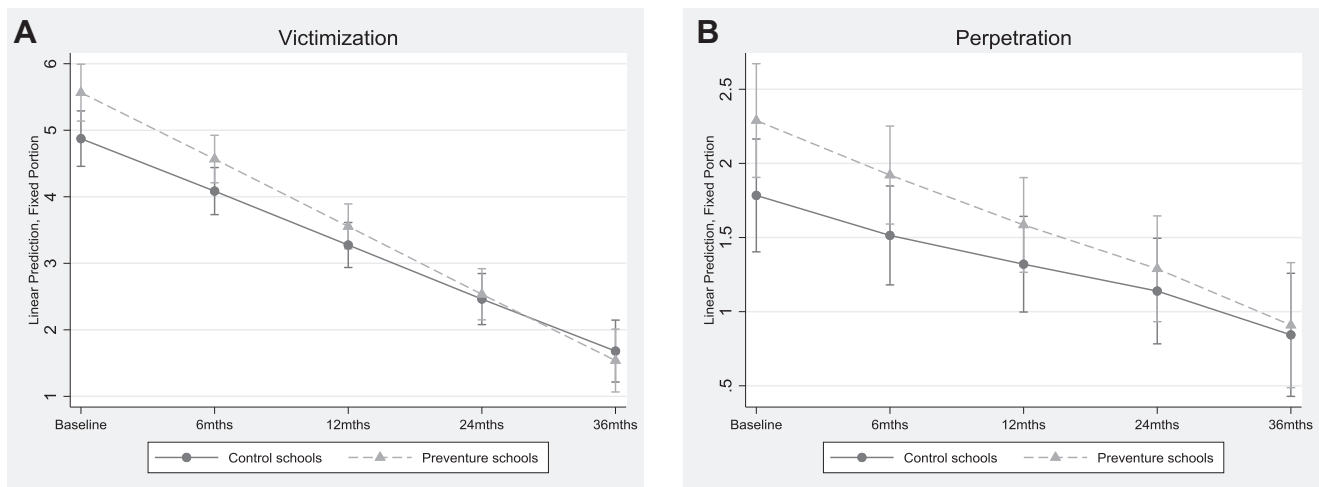
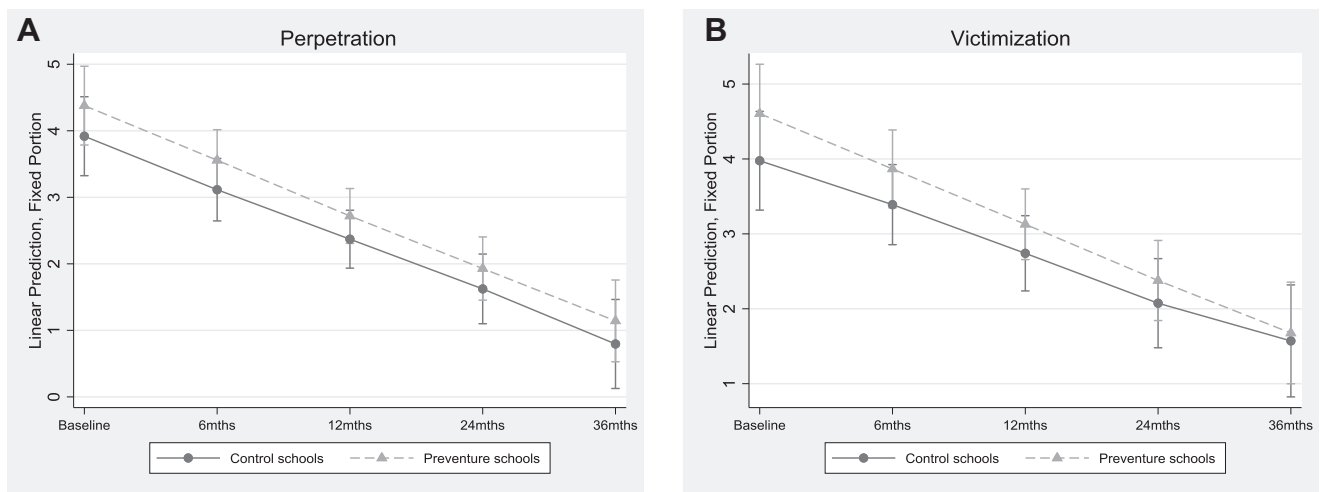
FIGURE 1 Victimization (A) and Perpetration (B) Scores Over Time Among High-Risk Victims in Preventure Versus Control Schools

FIGURE 2 Perpetration (A) and Victimization (B) Scores Over Time Among High-Risk Bullies in Preventure Versus Control Schools

This equates to a 25% greater reduction in conduct symptom scores among high-risk bullies in Preventure schools compared with high-risk bullies in control schools and a 56% greater reduction in suicidal ideation scores among high-risk female bullies in Preventure schools compared with high-risk female bullies in control schools. These results were striking, considering the significant reductions in bullying and harms over the 3-year study period, after only two group sessions. The Preventure intervention not only reduced additional bullying victimization but also reduced many emotional and behavioral problems commonly seen among victims and bullies. Whereas schoolwide antibullying programs have been found to be somewhat effective in reducing bullying in schools, it is evident that a high number of adolescents will continue to be bullied. Therefore, it is of utmost importance that universal, schoolwide bullying programs are combined with targeted programs aimed at reducing further bullying among vulnerable adolescents as well as reducing psychological problems among those involved in

bullying.⁵ It should be noted that the Preventure intervention did not result in population changes in bullying; continued efforts to improve universal prevention of adolescent bullying are critical.¹⁴

The results of the current study showed that the high-risk personality traits targeted in Preventure were over-represented among victims and bullies. Screening for such traits is a potential method for identifying individuals at risk of experiencing harm as a result of bullying involvement. Such routine screening in schools is important, as adolescents are often reluctant to initiate help-seeking for bullying or related distress.⁴⁴ Furthermore, bullying during adolescence is often covert and can be difficult for parents and school personnel to detect.⁴⁵ This method is also appealing in targeting intervention for the most vulnerable adolescents, as not all adolescents involved in bullying experience harms as a result.²⁰ Once identified, vulnerable adolescents can be provided with a brief, evidence-based CBT intervention that is likely to be effective in reducing further bullying involvement, inoculating them against

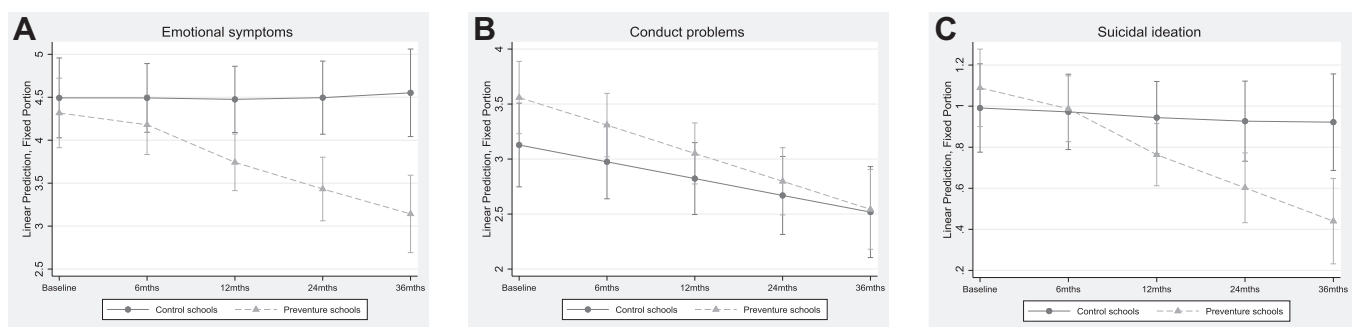
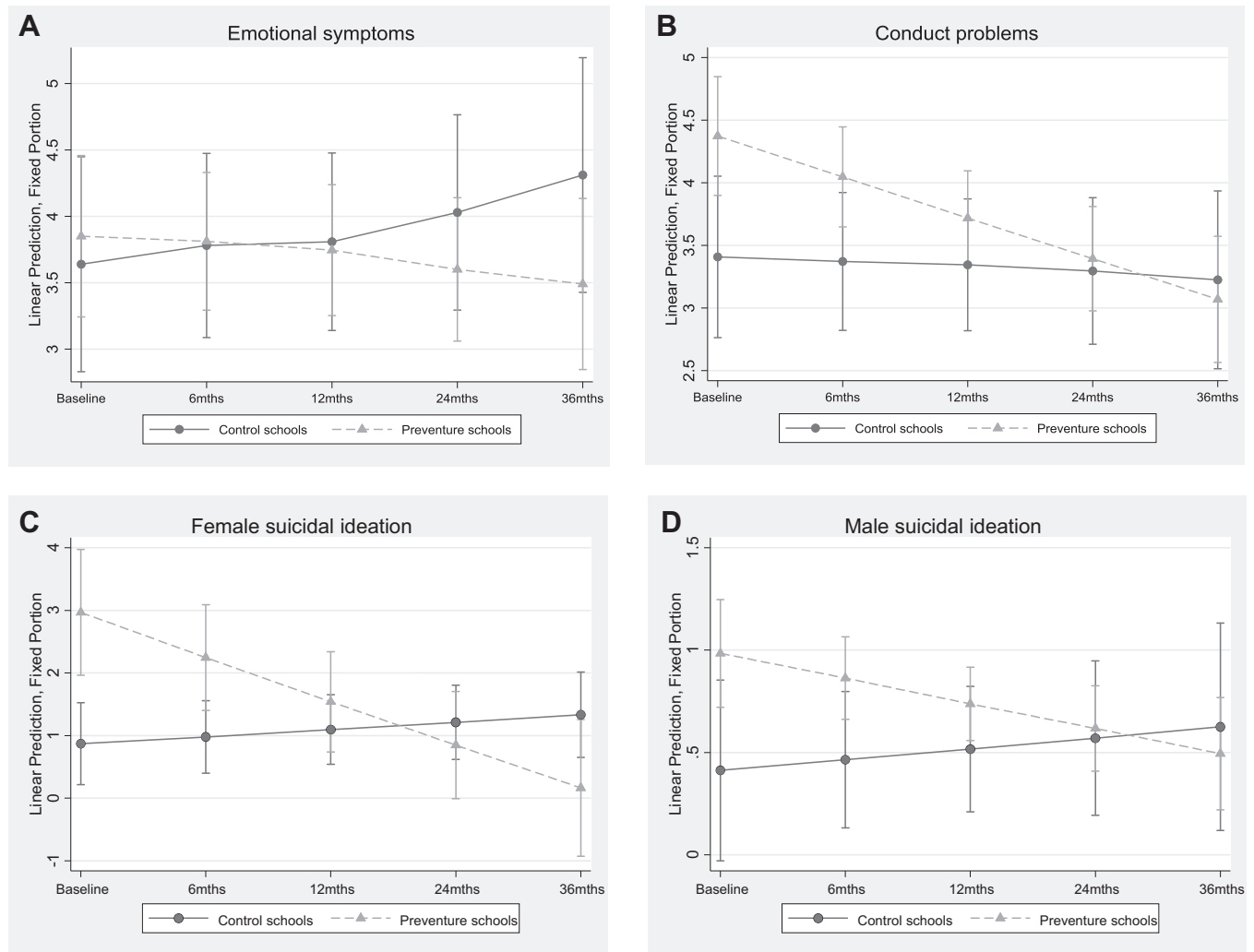
FIGURE 3 Emotional Symptoms (A), Conduct Problems (B), and Suicidal Ideation (C) Over Time Among High-Risk Victims in Preventure Versus Control Schools

FIGURE 4 Emotional Symptoms (A), Conduct Problems (B), and Suicidal Ideation (C, D) Over Time Among High-Risk Bullies in Preventure Versus Control Schools

experiencing psychological harm as a result of bullying involvement, and providing intervention for preexisting internalizing and externalizing problems common among adolescents involved in bullying.

While victimization and perpetration tend to decline over the secondary school years, some students experience chronic involvement in bullying, which results in greater harm. Universal bullying interventions alone are unlikely to be sufficient for such adolescents. As evident in Figures 1–4, even though victimization decreased over time for high-risk victims in both Preventure schools and control schools, emotional symptoms and suicidal ideation remained high in high-risk victims in control schools. Therefore, even if the bullying has ceased, it is not necessarily the case that the distress decreases among those previously victimized. More alarmingly, suicidal ideation and emotional symptoms appeared to increase over time among bullies in control

schools, highlighting an urgent need to intervene with this often-neglected group. It should be noted that suicidal ideation was significantly reduced for female high-risk bullies in Preventure schools compared with female high-risk bullies in control schools, but this trend was not significant for male high-risk bullies; this was possibly due to lower suicidal ideation in male high-risk bullies.

Overall, the results indicate that bullying involvement can be viewed as a marker for emotional and behavioral problems for victims and/or bullies and an avenue for providing early intervention for such problems, potentially preventing the onset of substance use and mental disorders.^{5,8} However, whereas victims of bullying are typically viewed as requiring some kind of psychological intervention, bullies are likely to be provided only with disciplinary action.⁴⁴ The results of this study show that bullies are often in need of intervention for emotional and behavioral

problems and can benefit from brief psychological intervention. However, there was no evidence that the Preventure intervention was successful in reducing bullying perpetration among the bullies. This may have been due to the low power to detect differences between the groups (numbers of high-risk victims and bullies over time in the Preventure and control conditions are provided in Table S2, available online); a larger sample size may demonstrate the effectiveness of Preventure for bullying perpetration. Alternatively, it is possible that a brief personality-targeted coping skills intervention is not sufficient for reducing bullying perpetration; for instance, a proportion of the bullies may have been motivated to continue using bullying as a means for achieving and maintaining social dominance.⁴⁶ Rather than a coping skills program, such bullies may require incentive to reduce bullying or assistance with using more prosocial methods for obtaining and maintaining a high social status within the peer group.^{5,47}

An important area of focus for bullying intervention is suicide risk, as evidenced by the current finding of frequent suicidal ideation among one in five victims and one in four bullies. Even though Preventure does not specifically address suicidal ideation, the focus on cognitive reappraisal and developing healthier ways of coping with distress was likely to have helped in reducing suicidal ideation. It is also possible that there were peer effects from being part of the groups, such as noting commonality between themselves and others and increased social support from other group members.

Cognitive restructuring has been highlighted as an essential element of indicated bullying interventions, as cognitions about one's bullying experiences contribute to the development of internalizing and externalizing problems.²⁶ Cognitive restructuring is a major component of the Preventure program; for instance, victims in the hopelessness group are prone to negative beliefs about themselves, others, and the future. Bullying victimization may confirm their negative beliefs about themselves (eg, "I am unlikely"), other people (eg, others cannot be trusted), and the future (eg, "I will never have any friends"), leading to withdrawal from others and, ultimately, depressive disorder. Learning to challenge these beliefs reduces the likelihood that a victim prone to negative thinking will develop a depressive disorder in response to the bullying. For adolescents high in impulsivity, bullying victimization can confirm beliefs about others always picking on them, resulting in antagonistic behavior, which can increase further victimization. Alternatively, bullying perpetration may result from the activation of a threat schema (eg, "everyone is out to get me"), which can exacerbate negative self-other beliefs (eg, "I'd better hurt them before they hurt

me"), leading to the young person becoming aggressive in social relationships to maintain power and control. Challenging negative beliefs about others could result in a reduction in bullying perpetration.

This study has some limitations that should be considered. It is possible that some participants did not report bullying perpetration, owing to socially desirable responding. The current study did not measure cyberbullying specifically, although research indicates that the problems among those involved in cyberbullying are similar to traditional bullying, so it is likely that the intervention could also be effective for cyberbullying.⁴⁸ The sample size was too small to examine bully-victims, categorical/clinical outcomes, or effects of the four personality groups separately. There were substantially more male participants in the Preventure schools than in the control schools; whereas the models take into account individual differences at baseline, it would be beneficial to examine the effects of Preventure on a larger sample with equal numbers of male and female participants to further evaluate its effects on bullying and associated harms. Intervention effects may have been inflated, owing to participants benefiting from the intervention more likely to stay in the study; however, such an effect should be minimal given the high follow-up rates (only 4% of the sample completed no follow-up surveys). The current study examined secondary effects of a substance use prevention program; the Preventure program does not specifically address bullying, although it often came up in student's personal examples in the group exercises and worksheets. Nonetheless, this study significantly adds to the bullying literature. A particular strength of this study was the longitudinal sample; few studies of bullying intervention have followed students over a 3-year period. This study also presents an innovative approach to bullying intervention, which achieved reductions in bullying victimization, emotional problems, conduct problems, and suicidal ideation. These problems are hugely important in the well-being of young people worldwide.

In conclusion, this study underscores the benefits of examining innovative methods for strengthening bullying intervention. The findings support the call for school-based bullying intervention to be holistic, accounting for the interrelatedness of bullying and other emotional and behavioral problems among adolescents.⁴⁹ It is evident that some level of bullying will remain in schools, so it is essential to arm those most vulnerable to bullying and its harms with effective coping skills. It is important to not only focus on improving the external environment of schools, but also to focus on strengthening resilience of young people to curb the harm caused by bullying.⁵⁰

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TABLE S1 Characteristics of Preventure and Control Schools at Baseline

	Control Schools	Preventure Schools
No. schools	13	13
No. participants	1,101	1,087
Gender, male, %	34.6	80.5
Age, y, mean (SD)	13.32 (.47)	13.34 (.49)
Personality classification, %		
NT	9.6	8.6
AS	11.4	13.8
IMP	10.4	9.5
SS	11.7	11.4
Victim, %	26.4	24.4
Bully, %	9.0	9.2
Bully and victim, %	5.8	6.3

Note: AS = anxiety sensitivity; IMP = impulsivity; NT = negative thinking; SS = sensation seeking.

TABLE S2 Bullying Over Time Among High-risk Victims and Bullies at Baseline

	High-risk Victims at Baseline (n = 296)		High-risk Bullies at Baseline (n = 130)	
	No. in Control Schools	No. in Preventure Schools	No. in Control Schools	No. in Preventure Schools
Victim status over time				
Baseline	153	143	41	44
6 mo	61	55	18	19
12 mo	58	45	24	15
24 mo	32	22	16	16
36 mo	29	16	8	8
Bully status over time				
Baseline	41	44	67	63
6 mo	19	18	18	12
12 mo	17	21	18	18
24 mo	11	9	12	12
36 mo	10	8	6	7

FIGURE S1 Participant Flow in the Climate and Preventure (CAP) Trial

Note: Originally produced in Newton *et al.*⁵¹ Distributed under the terms of the Creative Commons Attribution 4.0 International License <http://creativecommons.org/licenses/by/4.0/>. No changes were made to the original figure.

SUPPLMENTAL REFERENCE

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